

Nur Atiqah Binti Idris

Assignment 3

Questions for Participants (SQL Analysis) Basic Retrieval

1. Show all the data from the table.

```
17  -- Basic Retrieval
18  -- 1. Show all the data from the table.
19
20 •  SELECT * FROM cars;
21
22
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: |

	car_no	car_brand_name	car_status	listed	first_listing_date	sale_date	days
▶	Car No-1	Brand-A	Not Sold	TRUE	2021-06-20		
	Car No-2	Brand-B	Sold	TRUE	2020-08-11	2020-10...	80
	Car No-3	Brand-C	Not Sold	FALSE			
	Car No-4	Brand-C	Sold	TRUE	2020-08-13	2020-11...	109
	Car No-5	Brand-C	Not Sold	FALSE			
	Car No-6	Brand-C	Sold	TRUE	2020-08-26	2020-11...	79
	Car No-7	Brand-D	Sold	TRUE	2020-08-16	2020-12-...	85

cars 14 x

A full list of every car and its data

2. List all cars that are not sold.

```
21
22  -- 2. List all cars that are not sold.
23
24 •  SELECT * FROM cars
25      WHERE car_status = "Not Sold";
26
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: |

	car_no	car_brand_name	car_status	listed	first_listing_date	sale_date	days
▶	Car No-1	Brand-A	Not Sold	TRUE	2021-06-20		
	Car No-3	Brand-C	Not Sold	FALSE			
	Car No-5	Brand-C	Not Sold	FALSE			
	Car No-13	Brand-F	Not Sold	TRUE	2020-12-17		
	Car No-20	Brand-C	Not Sold	TRUE	2021-03-12		
	Car No-21	Brand-B	Not Sold	TRUE	2021-03-19		

cars 6 x

Result Grid
Form Editor
Read Only

Shows only cars that are still available or unsold.

3. List all cars from Brand-C.

```
28
29      -- 3. List all cars from Brand-C.
30
31  •      SELECT * FROM cars
32          WHERE car_brand_name = "Brand-C";
33
```

Result Grid



 Filter Rows:

Export: 

Wrap Cell Content: 

	car_no	car_brand_name	car_status	listed	first_listing_date	sale_date	days
▶	Car No-3	Brand-C	Not Sold	FALSE			
	Car No-4	Brand-C	Sold	TRUE	2020-08-13	2020-11...	109
	Car No-5	Brand-C	Not Sold	FALSE			
	Car No-6	Brand-C	Sold	TRUE	2020-08-26	2020-11...	79
	Car No-17	Brand-C	Sold	TRUE	2021-02-23	2021-04...	60
	Car No-18	Brand-C	Sold	TRUE	2021-03-03	2021-04...	56

Only cars made by Brand-C are shown.

4. Show the cars that have no first_listing_date.

```
47  -- 4. Show the cars that have no first_listing_date.
48
49  •  SELECT * FROM cars
50      WHERE listed = "FALSE";
51
52
53
```

Result Grid

Filter Rows:

Export:

Wrap Cell Content:

	car_no	car_brand_name	car_status	listed	first_listing_date	sale_date	days
▶	Car No-3	Brand-C	Not Sold	FALSE			
	Car No-5	Brand-C	Not Sold	FALSE			
	Car No-39	Brand-B	Not Sold	FALSE			
	Car No-40	Brand-K	Not Sold	FALSE			
	Car No-42	Brand-K	Not Sold	FALSE			
	Car No-77	Brand-B	Not Sold	FALSE			
	Car No-110	Brand-B	Not Sold	FALSE			

cars 13

Output

Lists cars that were never given a first listing date.

Filtering & Conditions

1. Find all cars that were listed but still not sold.

```
53  -- Filtering & Conditions
54  -- 1. Find all cars that were listed but still not sold.
55
56  •  SELECT * FROM cars
57      WHERE listed = "TRUE"
58      AND car_status = "Not Sold";
59
60
```

Result Grid | | Filter Rows: | Export: | Wrap Cell Content:

	car_no	car_brand_name	car_status	listed	first_listing_date	sale_date	days
▶	Car No-1	Brand-A	Not Sold	TRUE	2021-06-20		
	Car No-13	Brand-F	Not Sold	TRUE	2020-12-17		
	Car No-20	Brand-C	Not Sold	TRUE	2021-03-12		
	Car No-21	Brand-B	Not Sold	TRUE	2021-03-19		
	Car No-22	Brand-H	Not Sold	TRUE	2021-03-23		
	Car No-23	Brand-B	Not Sold	TRUE	2021-03-26		
	Car No-26	Brand-D	Not Sold	TRUE	2021-04-08		

cars 19 x

Cars that were actively listed but have not yet been purchased.

2. Show all cars that were sold within 60 days of listing (use DATEDIFF function).

```

60
61  -- 2. Show all cars that were sold within 60 days of listing (use DATEDIFF function).
62
63  • SELECT * FROM cars
64  WHERE car_status = "Sold"
65  AND datediff(sale_date, first_listing_date) <= 60;
66
67

```

	car_no	car_brand_name	car_status	listed	first_listing_date	sale_date	days
▶	Car No-10	Brand-E	Sold	TRUE	2020-10-25	2020-11...	32
	Car No-14	Brand-F	Sold	TRUE	2021-01-01	2021-01...	17
	Car No-16	Brand-F	Sold	TRUE	2021-02-06	2021-02...	20
	Car No-17	Brand-C	Sold	TRUE	2021-02-23	2021-04...	60
	Car No-18	Brand-C	Sold	TRUE	2021-03-03	2021-04...	56
	Car No-19	Brand-G	Sold	TRUE	2021-03-08	2021-04...	39
	Car No-24	Brand-C	Sold	TRUE	2021-03-24	2021-05...	50

Fast-selling cars — those that were sold within 60 days of being listed.

- Find all cars that were never listed (listed = false) but were sold.

```

68  -- 3. Find all cars that were never listed (listed = false) but were sold.
69
70  • SELECT * FROM cars
71  WHERE listed = "FALSE"
72  AND car_status = "Sold";

```

	car_no	car_brand_name	car_status	listed	first_listing_date	sale_date	days
▶	Car No-224	Brand-P	Sold	FALSE		2020-08...	44069
	Car No-633	Brand-D	Sold	FALSE		2020-11...	44161
	Car No-1087	Brand-D	Sold	FALSE		2020-12...	44186
	Car No-1174	Brand-G	Sold	FALSE		2021-04...	44312

Unusual cases where the car was sold without being officially listed — possibly internal transfers or data entry anomalies.

Aggregation & Insights

- Count how many cars have been sold.

```

74
75 -- Aggregation & Insights
76 -- 1. Count how many cars have been sold.
77
78 • SELECT COUNT(*) AS sold_cars_count
79 FROM cars
80 WHERE car_status = "Sold";
81

```

Result Grid	Filter Rows:	Export:
sold_cars_count		
607		

The total number of cars that have been sold.

- Find the earliest sale date and the latest sale date.

```

82
83 -- 2. Find the earliest sale date and the latest sale date.
84
85 • SELECT
86     MIN(sale_date) AS earliest_sale_date,
87     MAX(sale_date) AS latest_sale_date
88 FROM cars
89 WHERE car_status = "Sold";
90

```

Result Grid	Filter Rows:	Export:	Wrap Cell Content:
earliest_sale_date	latest_sale_date		
2020-08-24	2021-06-18		

Total number of cars per brand

- Show the total number of cars per car_brand_name.

```
91 -- 3. Show the total number of cars per car_brand_name.
92
93 • SELECT car_brand_name, COUNT(*) AS total_cars
94 FROM cars
95 GROUP BY car_brand_name
96 ORDER BY total_cars DESC;
97
98 -- 4. Show the number of sold and not sold cars per brand.
```

Result Grid |  Filter Rows: | Export:  | Wrap Cell Content: 

	car_brand_name	total_cars
▶	Brand-B	450
	Brand-C	266
	Brand-I	194
	Brand-D	136
	Brand-K	95
	Brand-J	55
	Brand-L	32

Result 24 x

Total number of cars per brand

4. Show the number of sold and not sold cars per brand.

```

98
99 -- 4. Show the number of sold and not sold cars per brand.
100
101 • SELECT
102     car_brand_name,
103     COUNT(*) AS total_cars,
104     COUNT(CASE WHEN car_status = 'Sold' THEN 1 END) AS sold_cars,
105     COUNT(CASE WHEN car_status = 'Not Sold' THEN 1 END) AS not_sold_cars
106 FROM cars
107 GROUP BY car_brand_name;
108

```

Result Grid |   Filter Rows: | Export:  | Wrap Cell Content: 

	car_brand_name	total_cars	sold_cars	not_sold_cars
▶	Brand-A	6	1	5
	Brand-B	450	207	243
	Brand-C	266	124	142
	Brand-D	136	59	77
	Brand-E	6	5	1
	Brand-F	16	8	8
	Brand-G	74	11	13

Number of sold and not sold cars per brand

Advanced Thinking

1. Which brand has the highest number of listings?

```
17
18 -- Advanced Thinking
19 -- 1. Which brand has the highest number of listings?
20
21 SELECT car_brand_name, COUNT(*) AS total_listings
22 FROM cars
23 WHERE listed = "TRUE"
24 GROUP BY car_brand_name
25 ORDER BY total_listings DESC
26 LIMIT 1;
```

Result Grid |   Filter Rows: | Export:  | Wrap Cell Content

car_brand_name	total_listings
Brand-B	423

Brand-B is the most cars listed for sale.

2. Which cars have missing data (null values) in any field?


```

128 -- 2. Which cars have missing data (null values) in any field?
129
130 • SELECT *
131 FROM cars
132 WHERE car_no = ""
133        OR car_brand_name = ""
134        OR car_status = ""
135        OR listed = ""
136        OR first_listing_date = ""
137        OR sale_date = ""
138        OR days = "";

```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

car_no	car_brand_name	car_status	listed	first_listing_date	sale_date	days
Car No-1	Brand-A	Not Sold	TRUE	2021-06-20		
Car No-3	Brand-C	Not Sold	FALSE			
Car No-5	Brand-C	Not Sold	FALSE			
Car No-13	Brand-F	Not Sold	TRUE	2020-12-17		
Car No-20	Brand-C	Not Sold	TRUE	2021-03-12		
Car No-21	Brand-B	Not Sold	TRUE	2021-03-19		
Car No-22	Brand-H	Not Sold	TRUE	2021-03-23		

Returns all cars that are incomplete in any way — either missing a sale date, brand name, listing flag, etc.

- Find the average time taken to sell a car (difference between first_listing_date and sale_date) only for sold cars.

```

141 -- 3. Find the average time taken to sell a car (difference between first_listing_date
142
143 • SELECT AVG(DATEDIFF(sale_date, first_listing_date)) AS avg_days_to_sell
144 FROM cars
145 WHERE car_status = "Sold"
146        AND sale_date IS NOT NULL
147        AND first_listing_date IS NOT NULL;

```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

avg_days_to_sell
69.8159

69.8159 is the average number of days it takes to sell a listed car.